

Effects of low habitual cocoa intake on blood pressure and bioactive nitric oxide: a randomized controlled trial.



CONTEXT: Regular intake of cocoa-containing foods is linked to lower cardiovascular mortality in observational studies. Short-term interventions of at most 2 weeks indicate that high doses of cocoa can improve endothelial function and reduce blood pressure (BP) due to the action of the cocoa polyphenols, but the clinical effect of low habitual cocoa intake on BP and the underlying BP-lowering mechanisms are unclear. OBJECTIVE: To determine effects of low doses of polyphenol-rich dark chocolate on BP. DESIGN, SETTING, AND PARTICIPANTS: Randomized, controlled, investigator-blinded, parallel-group trial involving 44 adults aged 56 through 73 years (24 women, 20 men) with untreated upper-range prehypertension or stage 1 hypertension without concomitant risk factors. The trial was conducted at a primary care clinic in Germany between January 2005 and December 2006. INTERVENTION: Participants were randomly assigned to receive for 18 weeks either 6.3 g (30 kcal) per day of dark chocolate containing 30 mg of polyphenols or matching polyphenol-free white chocolate. MAIN OUTCOME MEASURES: Primary outcome measure was the change in BP after 18 weeks. Secondary outcome measures were changes in plasma markers of vasodilative nitric oxide (S-nitrosoglutathione) and oxidative stress (8-isoprostane), and bioavailability of cocoa polyphenols. RESULTS: From baseline to 18 weeks, dark chocolate intake reduced mean (SD) systolic BP by -2.9 (1.6) mm Hg ($P < .001$) and diastolic BP by -1.9 (1.0) mm Hg ($P < .001$) without changes in body weight, plasma levels of lipids, glucose, and 8-isoprostane. Hypertension prevalence declined from 86% to 68%. The BP decrease was accompanied by a sustained increase of S-nitrosoglutathione by 0.23 (0.12) nmol/L ($P < .001$), and a dark chocolate dose resulted in the appearance of cocoa phenols in plasma. White chocolate intake caused no changes in BP or plasma biomarkers. CONCLUSIONS: Data in this relatively small sample of otherwise healthy individuals with above-optimal BP indicate that inclusion of small amounts of polyphenol-rich dark chocolate as part of a usual diet efficiently reduced BP and improved formation of vasodilative nitric oxide. TRIAL REGISTRATION: clinicaltrials.gov Identifier: NCT00421499.